Input Form (IF)

Score: 1/5 — Serious concern | Confidence: high

***Benchmark:** CRISISLTLSUM* | ***Context:** Ecuador — Humanitarian Crisis Social Media Analysts*

Input Form definition: Whether the input signal encoding (text, audio, image parameters) matches deployment conditions.

> Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

The benchmark's preprocessing pipeline is English-specific throughout: AllenNLP NER trained on CoNLL03 [Q33], a BERT-based NER fine-tuned on English Twitter [Q35], OpenStreetMap location matching calibrated to US/Australian geography [Q36], English lemmatization and lowercasing [Q129], and English keyword matching [Q130]. The authors explicitly acknowledge: 'replicating the same process for other language or based on other locations may be hard because of such dependencies' [Q119]. The Ecuador deployment requires processing Ecuadorian Spanish with informal register, code-switching with Kichwa, non-standard orthography (the 2008 unified Kichwa alphabet is contested [WEB-1]), and Kichwa/Shuar toponyms for which no NLP resources exist [WEB-1, WEB-12]. No Kichwa pre-trained model or annotated corpus has been identified [domain_specific_notes]. BERT/BART tokenization is English-calibrated [Q86, Q141, Q143]. This is a fundamental external-validity violation: the entire signal-processing chain is built on assumptions that do not hold in the deployment.

ID	Evidence
Q33	"We use three different modules. First, we use a pre-trained neural model from AllenNLP (Gardner et al., 2018), trained on CoNLL03 (Tjong Kim Sang and De Meulder, 2003), to extract entities with types of people, location, and organization." (p.3)
Q35	"Since location mentions are crucial in extracting local events and existing models have low performance detecting them from noisy tweets text, we further developed a BERT-based NER model tuned on Twitter data to detect location mentions." (p.4)
Q36	"Location Augmentation We use OpenStreetMap API to map location mentions to physical addresses." (p.4)
Q86	"For our first sentence-level classification model, we fine-tune BERT (Devlin et al., 2019) on the tweet sequences where every training example would have the form of "t0 [SEP] si"; where si ∈ S." (p.7)
Q119	"Accordingly, replicating the same process for other language or based on other locations may be hard because of such dependencies." (p.10)
Q129	"To avoid making a long list of keywords and ensure that different lexical formats of the same word are still considered in the keyword matching process, we maintain both a lower-cased and a lemittized version of the keywords and the tweet's text." (p.13)
Q130	"If any of the keywords exist in the text and in any of these formats, then the tweet is considered related to the category." (p.13)
Q141	"We fine-tune BERT base uncased on a single GPU for 10 epochs with a learning rate of 5e-5, batch size of 32, a seed of 42, and a maximum sequence length of 512." (p.14)
Q143	"We fine-tune BART based on a single GPU for ten epochs with a learning rate of 5e-5, batch size of 16, a seed of 42, and a maximum target sequence length of 512." (p.14)
WEB-1	Ecuadorian indigenous languages have official alphabets but written use outside schools is minimal; 2008 unified Kichwa orthography contested (https://books.openbookpublishers.com/10.11647/obp.0032/chapter11.html)
WEB-12	Approximately 527,333 Kichwa speakers and 59,894 Shuar speakers per INEC 2010 (https://en.wikipedia.org/wiki/Demographics_of_Ecuador)

Strengths

- The benchmark transparently flags the language-locality dependency [Q119], giving downstream users a clear signal that the pipeline is not portable; and the modular architecture (NER → location augmentation → clustering → classification) [Q28] provides a re-implementable template if components are swapped for Spanish/Kichwa-aware equivalents.

ID	Evidence

Q28	"Figure 2 shows the process for generating a set of noisy timelines starting from the Twitter stream and followed by pre-processing and knowledge enhancement steps, the online clustering method, and post-processing & cleaning steps." (p.3)
Q119	"Accordingly, replicating the same process for other language or based on other locations may be hard because of such dependencies." (p.10)

Checklist

Checklist item definitions:

ID	Assessment Question
IF-1	Compare signal distributions between source and target
IF-2	Check if regional infrastructure supports data capture specs
IF-3	Identify domain-specific form differences
IF-4	Document form mismatches harming external validity

Checklist responses:

IF-1: Signal distributions differ substantially: benchmark is English Twitter with US/Australian place-name density; deployment is Ecuadorian Spanish with informal register, crisis-specific Spanish lexicon ('damnificados', 'albergue', 'desbordamiento', 'lahares'), and Kichwa/Shuar toponyms.

IF-2: Ecuador has high mobile broadband (95.8% of mobile connections [WEB-15]) and ~77% national internet penetration [WEB-14], so Twitter/X capture is technically supported, but rural Oriente coverage is severely limited [infrastructure_notes]. The OSM location-augmentation step has unverified coverage quality for rural Ecuadorian parroquias.

IF-3: Domain-specific form differences: Spanish-tokenization-aware NER is required; Kichwa toponym recognition is not supported by any documented NLP tool; the benchmark's CoNLL03-trained NER [Q33] will systematically miss Spanish entity mentions and Kichwa place names.

IF-4: Documented: full pipeline mismatch on language, NER, location resolution, and tokenization. The authors' own limitation statement [Q119] concedes non-portability.

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WEB-1	Ecuadorian indigenous languages have official alphabets but written use outside schools is minimal; 2008 unified Kichwa orthography contested (https://books.openbookpublishers.com/10.11647/obp.0032/chapter11.html)
WEB-13	First Ecuadorian Spanish NLP corpus exists (2017) but does not cover Kichwa (https://ieeexplore.ieee.org/document/7889589)
WEB-14	Ecuador internet penetration ~77.2% (2024) (https://tradingeconomics.com/ecuador/individuals-using-the-internet-percent-of-population-wb-data.html)
WEB-15	95.8% of Ecuador mobile connections on 3G/4G/5G as of end 2025 (https://datareportal.com/reports/digital-2026-ecuador)

Information Gaps

- Quality of OpenStreetMap coverage for Ecuadorian rural parroquias and Kichwa community names is not assessed in available sources.

Requires Expert Verification

- Whether existing Spanish BERT variants combined with UrbangEnCy fine-tuning are operationally adequate for Ecuadorian crisis Twitter, and what resources would be needed to extend to Kichwa toponym recognition.

Remediation

Gap 1: Preprocessing pipeline (CoNLL03 NER, English BERT, OSM-on-US-geography, English lemmatization) is English-locked; the authors themselves flag non-portability [Q119].

Recommendation: Replace NER with Spanish-tuned models (e.g., Spanish BERT variants); add a Kichwa/Shuar toponym recognizer or gazetteer; validate OSM coverage on rural Ecuadorian parroquias and supplement with INEC administrative-unit databases.

ID	Evidence
Q119	“Accordingly, replicating the same process for other language or based on other locations may be hard because of such dependencies.” (p.10)